RAJALAKSHMI ENGINEERING COLLEGE

Kishore Kumar S (dob 20.03.2007) 251001123-2025-IT

K2

Started on	Wednesday, 19 August 2026, 9:05 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 3:14 PM
Time taken	42 days 6 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int binarySearch(int arr[], int left, int right, int target) {
    if (left > right)
        return -1;
    int mid = left + (right - left) / 2;
    if (arr[mid] == target)
        return mid;
    if (arr[mid] > target)
        return binarySearch(arr, left, mid - 1, target);
    return binarySearch(arr, mid + 1, right, target);
}
int findPair(int arr[], int left, int right, int x, int n) {
    if (left >= n - 1)
        return 0;
    int target = x - arr[left];
    int pos = binarySearch(arr, left + 1, n - 1, target);
    if (pos != -1) {
        printf("%d\n", arr[left]);
        printf("%d\n", arr[pos]);
        return 1;
    }
    return findPair(arr, left + 1, right, x, n);
}
int main() {
    int n, x;
    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    scanf("%d", &x);
    if (!findPair(arr, 0, n - 1, x, n))
        printf("No\n");
    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔

	Input	Expected	Got	
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course